Solution Architecture

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	5 Marks

End-to-End System Architecture:

VoyageAI is architected as a decoupled, multi-tier cloud application ensuring high throughput, resilience, and horizontal scalability.

Layer	Technologies	Architectural Responsibility
Client Tier	React 19, Vite, Tailwind v4, Zustand	Single Page Application hosted on Vercel Edge CDN with client-side SPA routing.
API Gateway Tier	FastAPI (Python 3.12), Uvicorn	Asynchronous REST endpoints hosted on Render with CORS regex matching.
AI Inference Tier	Groq Cloud LPU (openai/gpt-oss-120b)	Ultra-low latency LLM execution generating structured JSON itineraries.
Database Tier	Supabase PostgreSQL (Port 6543)	Transaction-pooled relational database with 7 tables, foreign keys, and RLS.
Security Tier	Python-Jose, Supabase Auth	ES256 asymmetric JWT verification and user_id session guards.